

ARK-491 — Authority Engine · Cost at Scale

Status: PREREGISTRATION (locked before execution)

Experiment Date: 2026-07-18

Series: ExecutionProof P02 Latency/Throughput/Scale

Question

What is the marginal compute cost per authority decision (and per million decisions) of the reference **Authority Engine** under a realistic running-service deployment model?

Analogous to ARK-486 (Verification Decision cost), applied to the current-state authority check. ARK-486 established that a per-request serverless price per in-process decision is a **category error** (the engine is an in-process library call executing millions of decisions/second, not a per-request Lambda invocation). ARK-491 adopts the same corrected cost basis.

Scope & Honesty Boundary

Minimal in-process reference `AuthorityEngine`, built to MEASURE, not the production engine. Cost figures are compute-time estimates for the reference implementation, not a customer quote.

Cost Model

Verdict basis — Scenario B (realistic running service): the engine runs inside an always-on service; marginal cost per decision = compute time × vCPU-second price.

- AWS Fargate: **\$0.04048 / vCPU-hour = \$0.00001124 / vCPU-second**

- $\text{cost_per_decision} = \text{vCPU-second price} \div \text{throughput (dec/s)}$

Disclosed upper bound — Scenario A (serverless-naive): if (incorrectly) billed as one serverless request per decision (~\$0.20/M + duration), reported for transparency only. NOT the verdict basis (see ARK-486 rationale).

Hypothesis

Scenario B cost $\leq$ **\$0.01 per million decisions**.

Component Under Test

Reference `AuthorityEngine` (exact, fail-closed). `engine/perf_harness.py --dimension cost` (derives cost from a 10s burst throughput measurement).

Methodology

- Measure burst throughput (10s), derive $\text{cost_per_decision} = \text{Fargate vCPU-second} \div \text{throughput}$
- Report both cost_per_decision and $\text{cost_per_million_decisions}$ (Scenario B), plus the Scenario A naive bound for disclosure

Metrics

Primary: `cost_per_million_decisions` (Scenario B)

Secondary: `cost_per_decision`, `throughput_dps`, `naive_serverless_per_million` (Scenario A), `accuracy`

Thresholds (Gate Conditions)

C1 (Cost): Scenario B `cost_per_million_decisions` $\leq$ \$0.01

C2 (Correctness gate): `valid`→ALLOW, `mutated`→DENY, `revoked`→DENY

Verdict: PASS if $C1 \wedge C2$; otherwise FAIL

Honest Findings Commitment

Both the realistic (Scenario B) and naive-serverless (Scenario A) figures are disclosed. The verdict is taken on Scenario B; the Scenario A bound is reported so readers can see the worst-case framing.

Execution Protocol

1. **Lock** prereg + MANIFEST.txt hashes
 2. **Execute** `engine/perf_harness.py --dimension cost`
 3. **Record** to `results/cost_results.json`
 4. **Evaluate** against C1, C2
 5. **Report** in RESULTS.md
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Investigator: Remnant Fieldworks Inc. / ExecutionProof Research Program

Covenant: RF Standing Covenant (preserve outcomes, bound claims, honest disclosure)